

INTEGRATED PROJECT

Efficiency analysis of port terminals using data from ANTAQ (the Brazilian National Waterway Transportation Agency)



WHAT IS IT?

- Teaching methodology developed by FATEC - São José dos Campos
- Initially implemented in 2019 in the Information Technology (IT) programs
- Currently in use across all programs of the institution

PILLARS

- **Real Problem-Based Learning:** Real-world challenges developed in collaboration with companies or organizations (academic partners)
- **External Validation:** Academic Partners oversee the process from conception to delivery and validate the relevance of the project
- **Agile Mindset:** Agile project management methodology (Scrum)

PROJECT OBJECTIVES



Business Intelligence

Develop a Business Intelligence (BI) platform to analyze port efficiency in Brazilian ports.



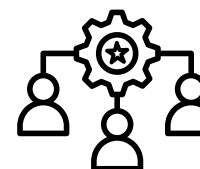
Processing Time

Provide information on processing times in port operations using data from ANTAQ



Productivity Comparison

Analyze and compare the productivity of port terminals, considering installed infrastructure and cargo volume handled.



Operations Time

Evaluate operation times, identifying which processes have longer waiting times and which are more efficient.



Operations Time

Investigate the main reasons for port stoppages and whether there is any seasonal correlation throughout the year.

LEAN FLOW

Smart Solutions for Logistics Challenges

TEAM MEMBERS



Arthur

Team DEV



Enzo

Team DEV



Ludmila

Team DEV



Vinicius

Product Owner



Yasmim

Scrum Master

WHAT WE HAVE LEARNED WITH THE PROJECT

Arthur

- Communication and integration of Jira and Github
- Construction of a Backlog on Jira Software
- Assistance in preparing the report

Enzo

- Responsible for preparing the overall report for this project.

Ludmila

- Construction of clear dashboards oriented toward decision-making.
- Communication and integration between teams (data, BI, documentation).
- Proper documentation for GitHub and academic presentations.

Vinicius

- Use the ANTAQ website and data
- Work more deeply with Python for data analysis and solve more complex data analysis problems
- Apply DEA models in various tools (Excel and RStudio)

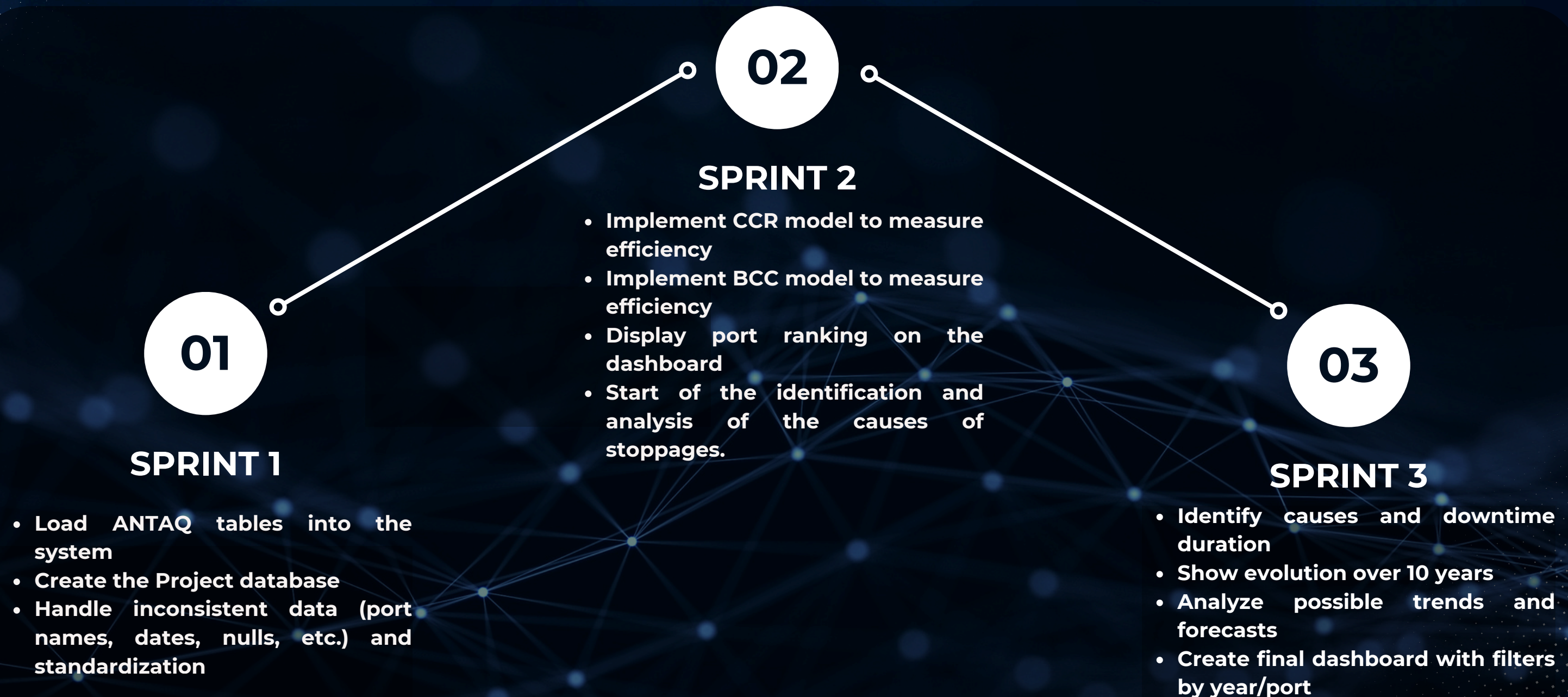
Yasmim

- File Manipulation and Process Automation
- Applying methods to inspect, filter, and understand the data before more complex analyses.
- Data Structuring for Logistics Analysis

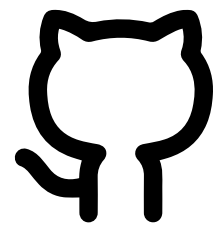
PRODUCT BACKLOG

UserStories	Description	Priority
US1	As an analyst, I want to import ANTAQ's raw data into the database to start the analysis.	Concluded SP1
US2	As an analyst, I want to clean, process, and standardize inconsistent data.	Concluded SP1
US3	As a manager, I want to calculate port and port complex efficiency using DEA CCR.	Concluded SP2
US4	As a manager, I want to calculate port efficiency using DEA BCC.	Concluded SP2
US5	As an analyst, I want to view a ranking of port efficiency.	Concluded SP2
US6	As a manager, I want to view the reasons and duration of stoppages.	Concluded SP3
US7	As a user, I want to filter data by port and causes	Concluded SP3

PRODUCT BACKLOG BY SPRINTS



PROJECT MANAGEMENT AND DOCUMENTATION TOOLS USED



GitHub



JiraSoftware



Whatsapp

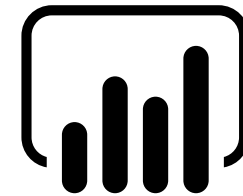
OTHER DIGITAL TOOLS USED



Excel



Python
(via Google Colab)

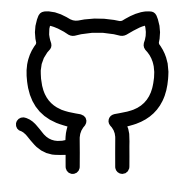


Power BI



RStudio

OUR PROJECT



GitHub



JiraSoftware

The background is a dark blue field filled with dynamic, flowing patterns of light. Multiple curved, concentric bands of bright blue and orange particles sweep across the frame, creating a sense of motion and depth. In the bottom right corner, there is a dense, white halftone dot pattern that fades into the dark background.

THANK YOU!